

## Verification, Validation, and Calibration of SDLC Survey Links

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### 1. Introduction

Verification and Validation (V&V) are essential quality control processes used to ensure that a system or instrument is both accurate and suitable for its intended use.

- **Verification** checks whether the evaluation process is applied correctly and consistently.
- **Validation** ensures that the evaluation actually measures what it is supposed to measure.
- **Calibration** involves adjusting the evaluation method when inconsistencies or issues are discovered.

In this assignment, three SDLC survey links (Basic, Normal, Advance) were evaluated over five rounds using a structured evaluation matrix. The goal was to improve the reliability and validity of the evaluation through iterative calibration.

### 2. Methodology

A fixed evaluation matrix was created and applied to all three links across five rounds.

#### Evaluation Criteria (Scale: 1-5)

- **Language:** clarity, readability, and appropriateness of wording
- **Quality:** organization, completeness, and usefulness of content
- **Performance:** speed, responsiveness, and usability

Each link was evaluated five times. After each round:

- Scores were recorded
- Justifications were written

- Any inconsistency triggered calibration and was documented in the logbook

### 3. Evaluation Results

#### Link 1 (Basic)

| Round | Language | Quality | Performance |
|-------|----------|---------|-------------|
| 1     | 4        | 3       | 4           |
| 2     | 4        | 3       | 4           |
| 3     | 5        | 3       | 4           |
| 4     | 4        | 3       | 4           |
| 5     | 4        | 3       | 4           |

#### Justification:

- **Language (4→5→4):** Clear overall, but some repetition and simple wording. Increased in round 3 after refining criteria.
- **Quality (3):** Basic level content, lacks depth and detail.
- **Performance (4):** Smooth and responsive with no major issues.

#### Interpretation:

The results are stable across rounds, indicating **high reliability**. Minor adjustment in round 3 improved **validity**.

#### Link 2 (Normal)

| Round | Language | Quality | Performance |
|-------|----------|---------|-------------|
| 1     | 3        | 4       | 4           |
| 2     | 3        | 4       | 4           |
| 3     | 4        | 4       | 4           |
| 4     | 4        | 4       | 4           |
| 5     | 4        | 4       | 4           |

#### Justification:

- **Language (3→4):** Initially slightly unclear, later adjusted after better understanding terminology.
- **Quality (4):** Well-structured and more complete than Link 1.
- **Performance (4):** Consistent and efficient.

#### Interpretation:

After calibration, scores became consistent, showing improved **reliability and validity**.

### Link 3 (Advance)

| Round | Language | Quality | Performance |
|-------|----------|---------|-------------|
| 1     | 4        | 5       | 3           |
| 2     | 4        | 5       | 3           |
| 3     | 4        | 5       | 4           |
| 4     | 4        | 5       | 4           |
| 5     | 4        | 5       | 4           |

#### Justification:

- **Language (4):** Clear but uses more advanced terminology.
- **Quality (5):** Highly detailed and comprehensive.
- **Performance (3→4):** Initially perceived slower, corrected after calibration.

#### Interpretation:

High **validity** due to strong content. Calibration improved performance evaluation consistency.

### 4. Adjustment Logbook (Calibration Record)

| Round | Link  | Adjustment                       | Reason                          | Impact                    |
|-------|-------|----------------------------------|---------------------------------|---------------------------|
| 2     | All   | Clarified Quality vs Language    | Overlap in criteria             | More accurate scoring     |
| 3     | Link1 | Adjusted Language scoring        | Better understanding of clarity | Slight score increase     |
| 3     | Link2 | Refined Language evaluation      | Terminology reconsidered        | Improved validity         |
| 3     | Link3 | Corrected Performance score      | Confused complexity with speed  | More accurate measurement |
| 4     | All   | Standardized evaluation approach | Reduce variation                | Increased reliability     |
| 5     | All   | No changes                       | Stability achieved              | High consistency          |

## 5. Final Analysis

Across all three links, the repeated evaluation process improved both reliability and validity:

- Early rounds showed slight inconsistencies due to unclear criteria
- Calibration helped refine the evaluation matrix
- Later rounds became stable and consistent

### Link Comparison:

- Link 1: Basic but consistent
- Link 2: Balanced and stable
- Link 3: Highest quality but initially misjudged performance

This demonstrates how calibration enhances evaluation accuracy.

## 6. Conclusion

This assignment demonstrated the importance of Verification, Validation, and Calibration:

- **Verification** ensured consistent use of the evaluation matrix
- **Validation** ensured meaningful and accurate measurement
- **Calibration** improved the matrix over time

By the final rounds, the evaluation process became both **reliable and valid**, fulfilling the

This demonstrates how iterative evaluation and calibration can significantly enhance the accuracy and consistency of measurement systems.